

**ILLINOIS INSTITUTE OF TECHNOLOGY  
DEPARTMENT OF COMPUTER SCIENCE**

**Database Organization**

**Course CS-425 02**

**Homework No.4**

**SQL Advanced window functions & OLAP queries**

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**Due Date:**

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**Summer 2024**

## Data input – question1

```
CREATE TABLE covid_statistics (  
    date DATE,  
    yyyymmdd INT,  
    state VARCHAR(2),  
    admissions INT,  
    current_num_in_hospital INT,  
    general INT,  
    high_care INT,  
    ICU INT,  
    isolation INT,  
    total_healthcare_workers_admitted INT,  
    num_discharged_alive INT,  
    hospital_deaths INT  
);
```

## Capture of the table

| InfoColumnsIndexesTriggersForeign keysPartitionsGrantsDDL |            |               |          |               |                |                                 |       |
|-----------------------------------------------------------|------------|---------------|----------|---------------|----------------|---------------------------------|-------|
| Column                                                    | Type       | Default Value | Nullable | Character Set | Collation      | Privileges                      | Extra |
| admissions                                                | int        |               | YES      |               |                | select,insert,update,references |       |
| current_num_in_h...                                       | int        |               | YES      |               |                | select,insert,update,references |       |
| date                                                      | date       |               | YES      |               |                | select,insert,update,references |       |
| general                                                   | int        |               | YES      |               |                | select,insert,update,references |       |
| high_care                                                 | int        |               | YES      |               |                | select,insert,update,references |       |
| hospital_deaths                                           | int        |               | YES      |               |                | select,insert,update,references |       |
| ICU                                                       | int        |               | YES      |               |                | select,insert,update,references |       |
| isolation                                                 | int        |               | YES      |               |                | select,insert,update,references |       |
| num_discharged_...                                        | int        |               | YES      |               |                | select,insert,update,references |       |
| state                                                     | varchar(2) |               | YES      | utf8mb4       | utf8mb4_090... | select,insert,update,references |       |
| total_healthcare_w...                                     | int        |               | YES      |               |                | select,insert,update,references |       |
| yyyymmdd                                                  | int        |               | YES      |               |                | select,insert,update,references |       |

| Indexes in Table |     |      |         |         | Index Details |  |
|------------------|-----|------|---------|---------|---------------|--|
| Visible          | Key | Type | Uniq... | Columns | Key Name:     |  |
|                  |     |      |         |         | Index Type:   |  |
|                  |     |      |         |         | Allows NULL:  |  |
|                  |     |      |         |         | Cardinality:  |  |
|                  |     |      |         |         | Comment:      |  |
|                  |     |      |         |         | User Comment: |  |

| Columns in table      |            |          |         |
|-----------------------|------------|----------|---------|
| Column                | Type       | Nullable | Indexes |
| date                  | date       | YES      |         |
| yyyymmdd              | int        | YES      |         |
| state                 | varchar(2) | YES      |         |
| admissions            | int        | YES      |         |
| current_num_in_h...   | int        | YES      |         |
| general               | int        | YES      |         |
| high_care             | int        | YES      |         |
| ICU                   | int        | YES      |         |
| isolation             | int        | YES      |         |
| total_healthcare_w... | int        | YES      |         |
| num_discharged_...    | int        | YES      |         |
| hospital_deaths       | int        | YES      |         |

| Column           | Datatype   | PK                                  | NN                                  | UQ                                  | B...                                | UN                                  | ZF                                  | AI                                  | G                                   | Default / Expression |
|------------------|------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|----------------------|
| date             | DATE       | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | NULL                 |
| yyyymmdd         | INT        | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | NULL                 |
| state            | VARCHAR(2) | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | NULL                 |
| admissions       | INT        | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | NULL                 |
| current_num...   | INT        | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | NULL                 |
| general          | INT        | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | NULL                 |
| high_care        | INT        | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | NULL                 |
| ICU              | INT        | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | NULL                 |
| isolation        | INT        | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | NULL                 |
| total_healthc... | INT        | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | NULL                 |
| num_dischar...   | INT        | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | NULL                 |
| hospital_deat... | INT        | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | NULL                 |
| <click to edit>  |            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |                      |

```

INSERT INTO covid_statistics (
    date, yyyymmdd, state, admissions, current_num_in_hospital, general, high_care, ICU, isolation, total_healthcare_workers_admitted,
    num_discharged_alive, hospital_deaths
) VALUES
('2020-05-24', 20200524, 'AL', 3046, 991, 715, 94, 151, 31, 165, 1615, 389),
('2020-05-25', 20200525, 'AZ', 3216, 1086, 805, 100, 146, 35, 156, 1669, 409),
('2020-05-26', 20200526, 'DE', 3439, 1139, 848, 96, 151, 44, 164, 1800, 447),
('2020-05-27', 20200527, 'DC', 3802, 1364, 1042, 119, 162, 41, 180, 1910, 472),
('2020-05-28', 20200528, 'FL', 4145, 1581, 1218, 119, 198, 46, 187, 2011, 497),
('2020-05-29', 20200529, 'DC', 4090, 1395, 1046, 117, 188, 44, 182, 2118, 519),
('2020-05-30', 20200530, 'FL', 4361, 1458, 1098, 115, 205, 40, 187, 2285, 559),
('2020-05-31', 20200531, 'DC', 4546, 1555, 1204, 108, 203, 40, 188, 2346, 587),
('2020-06-01', 20200601, 'AL', 4718, 1607, 1255, 101, 208, 43, 189, 2441, 608),
('2020-06-02', 20200602, 'AL', 4927, 1659, 1283, 115, 217, 44, 193, 2562, 643),
('2020-06-03', 20200603, 'AZ', 5078, 1638, 1258, 114, 219, 47, 195, 2713, 663),
('2020-06-04', 20200604, 'DE', 5377, 1717, 1316, 115, 242, 44, 204, 2893, 702),
('2020-06-06', 20200606, 'DC', 5945, 1916, 1456, 138, 265, 57, 210, 3175, 775),
('2020-06-07', 20200607, 'DE', 6180, 1928, 1466, 133, 268, 61, 214, 3348, 825),
('2020-06-08', 20200608, 'DC', 6353, 1974, 1500, 100, 300, 100, 215, 3446, 854),
('2020-06-09', 20200609, 'DE', 6682, 2054, 1540, 159, 286, 69, 219, 3628, 921),
('2020-06-10', 20200610, 'DC', 6992, 2113, 1583, 157, 301, 72, 227, 3836, 960),
('2020-06-11', 20200611, 'AZ', 7291, 2140, 1601, 169, 300, 70, 230, 4052, 1016),
('2020-06-12', 20200612, 'DE', 7556, 2287, 1709, 174, 326, 78, 233, 4148, 1035),
('2020-06-13', 20200613, 'AL', 7837, 2263, 1684, 176, 329, 71, 237, 4396, 1088),
('2020-06-14', 20200614, 'AZ', 8053, 2332, 1739, 183, 338, 72, 238, 4502, 1128),
('2020-06-15', 20200615, 'DE', 8292, 2468, 1869, 189, 340, 70, 239, 4576, 1156),
('2020-06-16', 20200616, 'DC', 8688, 2565, 1922, 216, 352, 75, 245, 4804, 1226),
('2020-06-17', 20200617, 'AL', 8902, 2661, 1983, 223, 380, 75, 249, 4878, 1265),
('2020-06-18', 20200618, 'AZ', 9321, 2741, 2008, 249, 398, 86, 264, 5147, 1333),
('2020-06-19', 20200619, 'DE', 9693, 2949, 2164, 263, 443, 79, 265, 5257, 1372),
('2020-06-20', 20200620, 'DC', 10020, 3005, 2163, 292, 468, 82, 272, 5479, 1420),
('2020-06-21', 20200621, 'DE', 10360, 3088, 2199, 315, 490, 84, 278, 5669, 1484),
('2020-06-22', 20200622, 'DC', 10700, 3263, 2348, 333, 499, 83, 290, 5805, 1515),
('2020-06-23', 20200623, 'AZ', 11320, 3501, 2529, 341, 538, 93, 302, 6096, 1592),
('2020-06-24', 20200624, 'FL', 11850, 3622, 2619, 371, 545, 87, 314, 6397, 1678),
('2020-06-25', 20200625, 'AZ', 12260, 3567, 2526, 376, 571, 94, 317, 6807, 1739),
('2020-06-27', 20200627, 'DE', 13850, 4051, 2861, 412, 645, 124, 357, 7682, 1966),

```

## Capture of data insertion

Homework4DB

covid\_statistics

Result Grid

Filter Rows:

Search

Export:

|  | date       | yyyymmdd | state | admissions | current_num_in_hosp... | general | high_care | ICU | isolation | total_healthcare_workers_admit... | num_discharged_ali... | hospital_deat... |
|--|------------|----------|-------|------------|------------------------|---------|-----------|-----|-----------|-----------------------------------|-----------------------|------------------|
|  | 2020-05-24 | 20200524 | AL    | 3046       | 991                    | 715     | 94        | 151 | 31        | 165                               | 1615                  | 389              |
|  | 2020-05-25 | 20200525 | AZ    | 3216       | 1086                   | 805     | 100       | 146 | 35        | 156                               | 1669                  | 409              |
|  | 2020-05-26 | 20200526 | DE    | 3439       | 1139                   | 848     | 96        | 151 | 44        | 164                               | 1800                  | 447              |
|  | 2020-05-27 | 20200527 | DC    | 3802       | 1364                   | 1042    | 119       | 162 | 41        | 180                               | 1910                  | 472              |
|  | 2020-05-28 | 20200528 | FL    | 4145       | 1581                   | 1218    | 119       | 198 | 46        | 187                               | 2011                  | 497              |
|  | 2020-05-29 | 20200529 | DC    | 4090       | 1395                   | 1046    | 117       | 188 | 44        | 182                               | 2118                  | 519              |
|  | 2020-05-30 | 20200530 | FL    | 4361       | 1458                   | 1098    | 115       | 205 | 40        | 187                               | 2285                  | 559              |
|  | 2020-05-31 | 20200531 | DC    | 4546       | 1555                   | 1204    | 108       | 203 | 40        | 188                               | 2346                  | 587              |
|  | 2020-06-01 | 20200601 | AL    | 4718       | 1607                   | 1255    | 101       | 208 | 43        | 189                               | 2441                  | 608              |
|  | 2020-06-02 | 20200602 | AL    | 4927       | 1659                   | 1283    | 115       | 217 | 44        | 193                               | 2562                  | 643              |
|  | 2020-06-03 | 20200603 | AZ    | 5078       | 1638                   | 1258    | 114       | 219 | 47        | 195                               | 2713                  | 663              |
|  | 2020-06-04 | 20200604 | DE    | 5377       | 1717                   | 1316    | 115       | 242 | 44        | 204                               | 2893                  | 702              |
|  | 2020-06-06 | 20200606 | DC    | 5945       | 1916                   | 1456    | 138       | 265 | 57        | 210                               | 3175                  | 775              |
|  | 2020-06-07 | 20200607 | DE    | 6180       | 1928                   | 1466    | 133       | 268 | 61        | 214                               | 3348                  | 825              |
|  | 2020-06-08 | 20200608 | DC    | 6353       | 1974                   | 1500    | 100       | 300 | 100       | 215                               | 3446                  | 854              |
|  | 2020-06-09 | 20200609 | DE    | 6682       | 2054                   | 1540    | 159       | 286 | 69        | 219                               | 3628                  | 921              |
|  | 2020-06-10 | 20200610 | DC    | 6992       | 2113                   | 1583    | 157       | 301 | 72        | 227                               | 3836                  | 960              |
|  | 2020-06-11 | 20200611 | AZ    | 7291       | 2140                   | 1601    | 169       | 300 | 70        | 230                               | 4052                  | 1016             |
|  | 2020-06-12 | 20200612 | DE    | 7556       | 2287                   | 1709    | 174       | 326 | 78        | 233                               | 4148                  | 1035             |
|  | 2020-06-13 | 20200613 | AL    | 7837       | 2263                   | 1684    | 176       | 329 | 71        | 237                               | 4396                  | 1088             |
|  | 2020-06-14 | 20200614 | AZ    | 8053       | 2332                   | 1739    | 183       | 338 | 72        | 238                               | 4502                  | 1128             |
|  | 2020-06-15 | 20200615 | DE    | 8292       | 2468                   | 1869    | 189       | 340 | 70        | 239                               | 4576                  | 1156             |
|  | 2020-06-16 | 20200616 | DC    | 8688       | 2565                   | 1922    | 216       | 352 | 75        | 245                               | 4804                  | 1226             |
|  | 2020-06-17 | 20200617 | AL    | 8902       | 2661                   | 1983    | 223       | 380 | 75        | 249                               | 4878                  | 1265             |
|  | 2020-06-18 | 20200618 | AZ    | 9321       | 2741                   | 2008    | 249       | 398 | 86        | 264                               | 5147                  | 1333             |
|  | 2020-06-19 | 20200619 | DE    | 9693       | 2949                   | 2164    | 263       | 443 | 79        | 265                               | 5257                  | 1372             |
|  | 2020-06-20 | 20200620 | DC    | 10020      | 3005                   | 2163    | 292       | 468 | 82        | 272                               | 5479                  | 1420             |
|  | 2020-06-21 | 20200621 | DE    | 10360      | 3088                   | 2199    | 315       | 490 | 84        | 278                               | 5669                  | 1484             |
|  | 2020-06-22 | 20200622 | DC    | 10700      | 3263                   | 2348    | 333       | 499 | 83        | 290                               | 5805                  | 1515             |
|  | 2020-06-23 | 20200623 | AZ    | 11320      | 3501                   | 2529    | 341       | 538 | 93        | 302                               | 6096                  | 1592             |
|  | 2020-06-24 | 20200624 | FL    | 11850      | 3622                   | 2619    | 371       | 545 | 87        | 314                               | 6397                  | 1678             |
|  | 2020-06-25 | 20200625 | AZ    | 12260      | 3567                   | 2526    | 376       | 571 | 94        | 317                               | 6807                  | 1739             |
|  | 2020-06-27 | 20200627 | DE    | 13850      | 4051                   | 2861    | 412       | 645 | 124       | 357                               | 7682                  | 1966             |
|  | 2020-06-28 | 20200628 | FL    | 14100      | 4150                   | 2941    | 421       | 668 | 120       | 365                               | 7797                  | 1995             |
|  | 2020-06-29 | 20200629 | DE    | 14560      | 4250                   | 3035    | 420       | 667 | 128       | 376                               | 8042                  | 2116             |
|  | 2020-06-30 | 20200630 | DC    | 15130      | 4418                   | 3160    | 452       | 690 | 116       | 394                               | 8364                  | 2189             |
|  | 2020-07-01 | 20200701 | FL    | 15370      | 4461                   | 3173    | 452       | 699 | 137       | 396                               | 8528                  | 2217             |
|  | 2020-07-02 | 20200702 | FL    | 15860      | 4616                   | 3313    | 464       | 708 | 131       | 402                               | 8805                  | 2273             |
|  | 2020-07-03 | 20200703 | DC    | 16740      | 5006                   | 3614    | 471       | 774 | 147       | 434                               | 9166                  | 2400             |

covid\_statistics 3

## Date input – question 2

```
create table regional_sales (
product char(100),
region char(100),
country char(100),
city char(100),
sales int;
```

Capture of the table

| Info Columns Indexes Triggers Foreign keys Partitions Grants DDL |           |               |          |               |                |                                 |     |
|------------------------------------------------------------------|-----------|---------------|----------|---------------|----------------|---------------------------------|-----|
| Column                                                           | Type      | Default Value | Nullable | Character Set | Collation      | Privileges                      | Ext |
| city                                                             | char(100) |               | YES      | utf8mb4       | utf8mb4_090... | select,insert,update,references |     |
| country                                                          | char(100) |               | YES      | utf8mb4       | utf8mb4_090... | select,insert,update,references |     |
| product                                                          | char(100) |               | YES      | utf8mb4       | utf8mb4_090... | select,insert,update,references |     |
| region                                                           | char(100) |               | YES      | utf8mb4       | utf8mb4_090... | select,insert,update,references |     |
| sales                                                            | int       |               | YES      |               |                | select,insert,update,references |     |

Info

Columns

Indexes

Triggers

Foreign ke

Indexes in Table

| Visible | Key | Type | Uniq... | Columns |
|---------|-----|------|---------|---------|
|         |     |      |         |         |
|         |     |      |         |         |
|         |     |      |         |         |
|         |     |      |         |         |

Index Details

Key Name:  
Index Type:  
Allows NULL:  
Cardinality:  
Comment:  
User Comment:

Columns in table

| Column  | Type      | Nullable | Indexes |
|---------|-----------|----------|---------|
| product | char(100) | YES      |         |
| region  | char(100) | YES      |         |
| country | char(100) | YES      |         |
| city    | char(100) | YES      |         |
| sales   | int       | YES      |         |

| Column          | Datatype  | PK | NN | UQ | B... | UN | ZF | AI | G | Default / Expression |
|-----------------|-----------|----|----|----|------|----|----|----|---|----------------------|
| product         | CHAR(100) |    |    |    |      |    |    |    |   | NULL                 |
| region          | CHAR(100) |    |    |    |      |    |    |    |   | NULL                 |
| country         | CHAR(100) |    |    |    |      |    |    |    |   | NULL                 |
| city            | CHAR(100) |    |    |    |      |    |    |    |   | NULL                 |
| sales           | INT       |    |    |    |      |    |    |    |   | NULL                 |
| <click to edit> |           |    |    |    |      |    |    |    |   |                      |

```
INSERT INTO regional_sales (Product, Region, Country, City, Sales) VALUES
('A', 'Asia', 'China', 'Beijing', 2500),
('A', 'Asia', 'China', 'Beijing', 1700),
('A', 'Asia', 'Japan', 'Tokyo', 800),
('A', 'North America', 'Canada', 'Toronto', 900),
('A', 'North America', 'USA', 'New York', 400),
('A', 'North America', 'Canada', 'Toronto', 2300),
('A', 'North America', 'USA', 'Chicago', 7800),
('A', 'Europe', 'Germany', 'Berlin', 300),
```

## Capture of data insertion

**Limit to 10000 rows**

```
SELECT * FROM Homework4DB.regional_sales;
```

100% 1:1

**Result Grid** Filter Rows: Search Export:

|                          | product | region        | country   | city       | sales |
|--------------------------|---------|---------------|-----------|------------|-------|
| <input type="checkbox"/> | A       | Asia          | China     | Beijing    | 2500  |
| <input type="checkbox"/> | A       | Asia          | China     | Beijing    | 1700  |
| <input type="checkbox"/> | A       | Asia          | Japan     | Tokyo      | 800   |
| <input type="checkbox"/> | A       | North America | Canada    | Toronto    | 900   |
| <input type="checkbox"/> | A       | North America | USA       | New York   | 400   |
| <input type="checkbox"/> | A       | North America | Canada    | Toronto    | 2300  |
| <input type="checkbox"/> | A       | North America | USA       | Chicago    | 7800  |
| <input type="checkbox"/> | A       | Europe        | Germany   | Berlin     | 300   |
| <input type="checkbox"/> | B       | Europe        | Germany   | Berlin     | 700   |
| <input type="checkbox"/> | B       | Europe        | France    | Paris      | 5500  |
| <input type="checkbox"/> | B       | Europe        | France    | Paris      | 2300  |
| <input type="checkbox"/> | B       | Europe        | France    | Paris      | 900   |
| <input type="checkbox"/> | B       | Europe        | Italy     | Rome       | 400   |
| <input type="checkbox"/> | B       | Australia     | Australia | Canberra   | 2500  |
| <input type="checkbox"/> | B       | Australia     | Australia | Canberra   | 3500  |
| <input type="checkbox"/> | B       | Australia     | New Ze... | Wellington | 5500  |

## Homework 1.4:

### Data input

```
CREATE TABLE employees (  
    row_num INT PRIMARY KEY,  
    first_name VARCHAR(50),  
    last_name VARCHAR(50),  
    salary int  
);
```

### Capture of the table

| Info Columns Indexes Triggers Foreign keys Partitions Grants DDL |             |               |          |               |                |                                 |       |
|------------------------------------------------------------------|-------------|---------------|----------|---------------|----------------|---------------------------------|-------|
| Column                                                           | Type        | Default Value | Nullable | Character Set | Collation      | Privileges                      | Extra |
| first_name                                                       | varchar(50) |               | YES      | utf8mb4       | utf8mb4_090... | select,insert,update,references |       |
| last_name                                                        | varchar(50) |               | YES      | utf8mb4       | utf8mb4_090... | select,insert,update,references |       |
| row_num                                                          | int         |               | NO       |               |                | select,insert,update,references |       |
| salary                                                           | int         |               | YES      |               |                | select,insert,update,references |       |

| Info Columns Indexes Triggers Foreign keys Partitions Grants DDL |             |          |         |         |               |        |
|------------------------------------------------------------------|-------------|----------|---------|---------|---------------|--------|
| Indexes in Table                                                 |             |          |         |         | Index Details |        |
| Visible                                                          | Key         | Type     | Uniq... | Columns | Key Name:     | Packed |
| <input checked="" type="checkbox"/>                              | PRIMARY     | BTREE    | YES     | row_num | Index Type:   | Unique |
|                                                                  |             |          |         |         | Allows NULL:  |        |
|                                                                  |             |          |         |         | Cardinality:  |        |
|                                                                  |             |          |         |         | Comment:      |        |
|                                                                  |             |          |         |         | User Comment: |        |
| Columns in table                                                 |             |          |         |         |               |        |
| Column                                                           | Type        | Nullable | Indexes |         |               |        |
| row_num                                                          | int         | NO       | PRIMARY |         |               |        |
| first_name                                                       | varchar(50) | YES      |         |         |               |        |
| last_name                                                        | varchar(50) | YES      |         |         |               |        |
| salary                                                           | int         | YES      |         |         |               |        |

| Column          | Datatype    | PK                                  | NN                                  | UQ                       | B...                     | UN                       | ZF                       | AI                       | G                        | Default / E: |
|-----------------|-------------|-------------------------------------|-------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------|
| row_num         | INT         | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |              |
| first_name      | VARCHAR(50) | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | NULL         |
| last_name       | VARCHAR(50) | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | NULL         |
| salary          | INT         | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | NULL         |
| <click to edit> |             |                                     |                                     |                          |                          |                          |                          |                          |                          |              |
|                 |             | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |              |

```
INSERT INTO employees (row_num, first_name, last_name, salary) VALUES
```

```
(1, 'Karen', 'Colmenares', 2500),
```

```
(2, 'Guy', 'Himuro', 2600),
```

```
(3, 'Irene', 'Mikklineni', 2700),
```

```
(4, 'Sigal', 'Tobias', 2800),
```

```
(5, 'Shelli', 'Baida', 2900),
```

```
(6, 'Alexander', 'Khoo', 3100),
```

```
(7, 'Britney', 'Everett', 3900),
```

```
(8, 'Sarah', 'Bell', 4000),
```

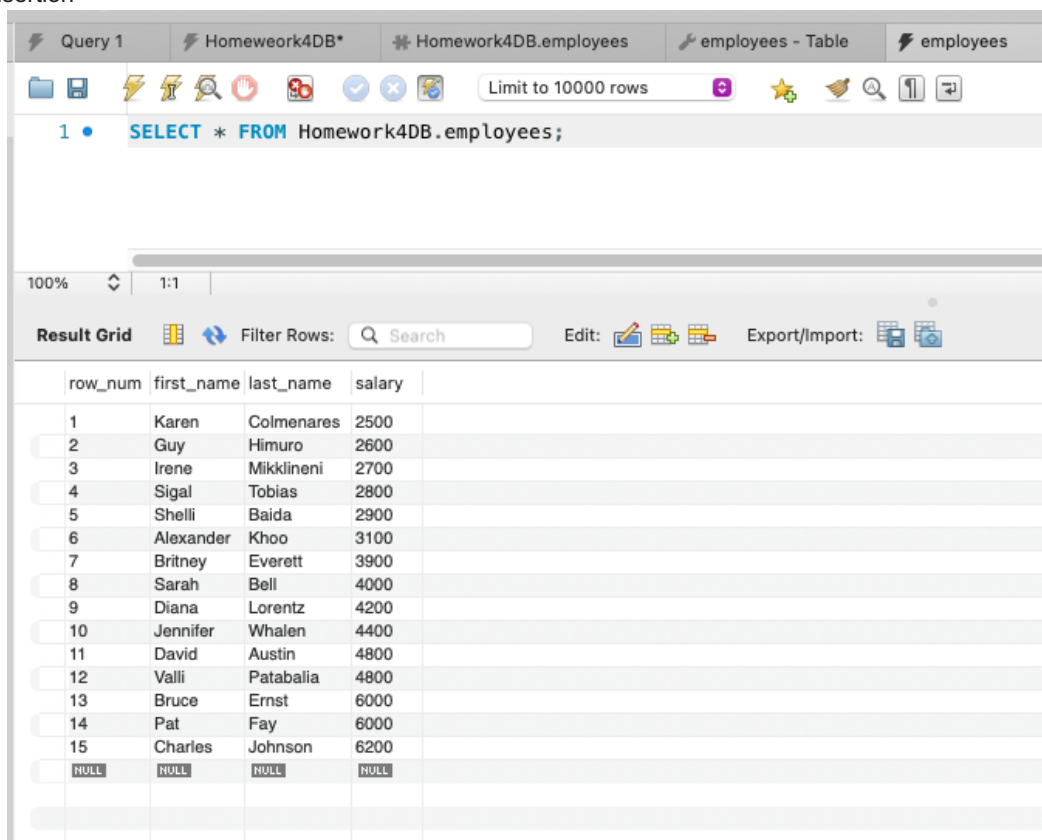
```
(9, 'Diana', 'Lorentz', 4200),
```

```
(10, 'Jennifer', 'Whalen', 4400),
```

```
(11, 'David', 'Austin', 4800),
```

(12, 'Valli', 'Patabalia', 4800),  
(13, 'Bruce', 'Ernst', 6000),  
(14, 'Pat', 'Fay', 6000),  
(15, 'Charles', 'Johnson', 6200);

Capture of data insertion



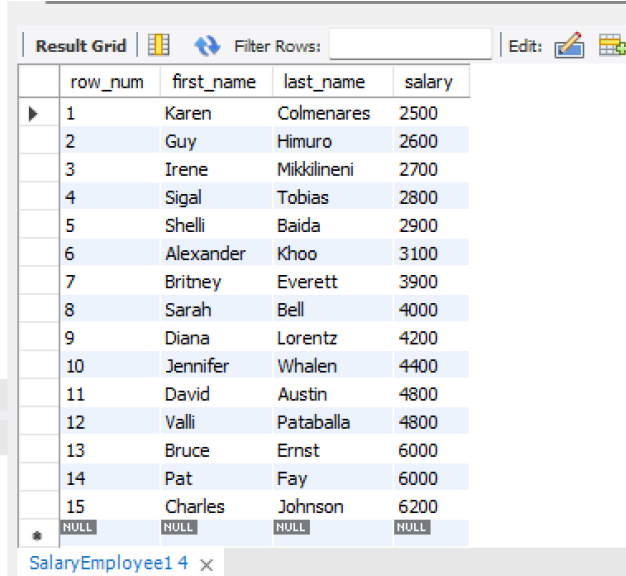
The screenshot shows a database management interface with a query editor at the top containing the SQL statement: `SELECT * FROM Homework4DB.employees;`. Below the editor is a 'Result Grid' displaying the data from the 'employees' table. The grid has four columns: 'row\_num', 'first\_name', 'last\_name', and 'salary'. It contains 15 rows of employee data, followed by a row with 'NULL' values. The interface includes various icons for file operations, a search bar, and options for editing and exporting data.

| row_num | first_name | last_name   | salary |
|---------|------------|-------------|--------|
| 1       | Karen      | Colmenares  | 2500   |
| 2       | Guy        | Himuro      | 2600   |
| 3       | Irene      | Mikkilineni | 2700   |
| 4       | Sigal      | Tobias      | 2800   |
| 5       | Shelli     | Baida       | 2900   |
| 6       | Alexander  | Khoo        | 3100   |
| 7       | Britney    | Everett     | 3900   |
| 8       | Sarah      | Bell        | 4000   |
| 9       | Diana      | Lorentz     | 4200   |
| 10      | Jennifer   | Whalen      | 4400   |
| 11      | David      | Austin      | 4800   |
| 12      | Valli      | Patabalia   | 4800   |
| 13      | Bruce      | Ernst       | 6000   |
| 14      | Pat        | Fay         | 6000   |
| 15      | Charles    | Johnson     | 6200   |
| NULL    | NULL       | NULL        | NULL   |



## PART I: advanced window functions

Using the dataset provided (employee) below, write the SQL queries, and screenshot the output for each of the window features given in the query (**Assume: Order by Salary ASC**).



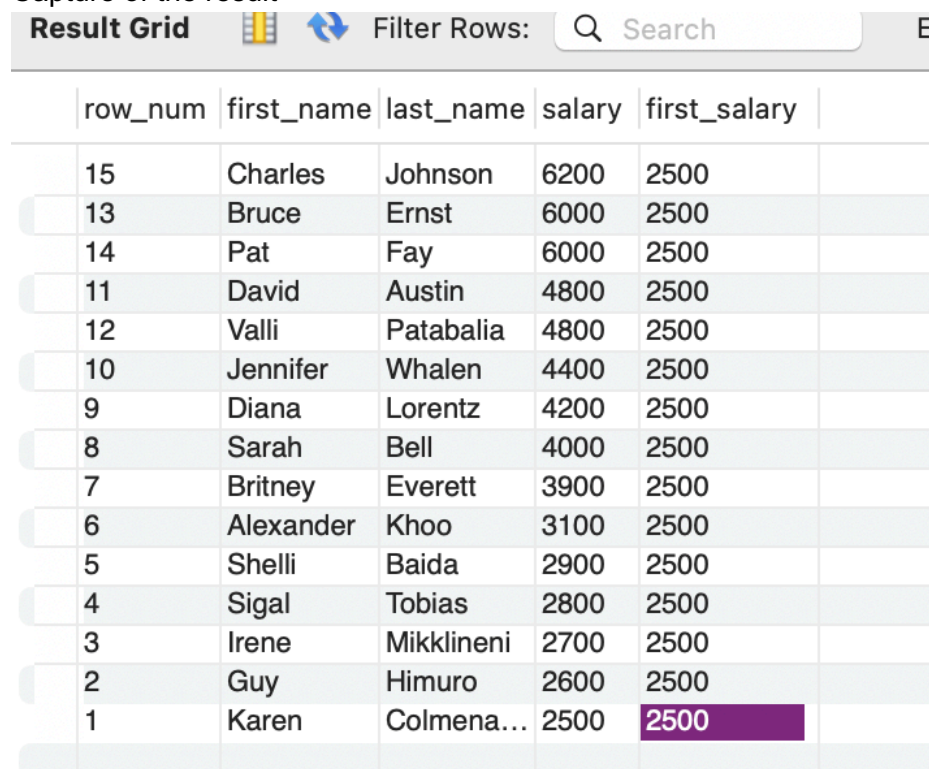
| row_num | first_name | last_name   | salary |
|---------|------------|-------------|--------|
| 1       | Karen      | Colmenares  | 2500   |
| 2       | Guy        | Himuro      | 2600   |
| 3       | Irene      | Mikkilineni | 2700   |
| 4       | Sigal      | Tobias      | 2800   |
| 5       | Shelli     | Baida       | 2900   |
| 6       | Alexander  | Khoo        | 3100   |
| 7       | Britney    | Everett     | 3900   |
| 8       | Sarah      | Bell        | 4000   |
| 9       | Diana      | Lorentz     | 4200   |
| 10      | Jennifer   | Whalen      | 4400   |
| 11      | David      | Austin      | 4800   |
| 12      | Valli      | Pataballa   | 4800   |
| 13      | Bruce      | Ernst       | 6000   |
| 14      | Pat        | Fay         | 6000   |
| 15      | Charles    | Johnson     | 6200   |
| *       | NULL       | NULL        | NULL   |

1. Write a query to compute for the **FIRST\_VALUE()** given the above dataset and return the value along with the entire row.

#QUESTION 1

```
SELECT *,
  FIRST_VALUE(salary) OVER (ORDER BY salary ASC) AS first_salary
FROM
  employees
ORDER BY
  Salary DESC;
```

Capture of the result



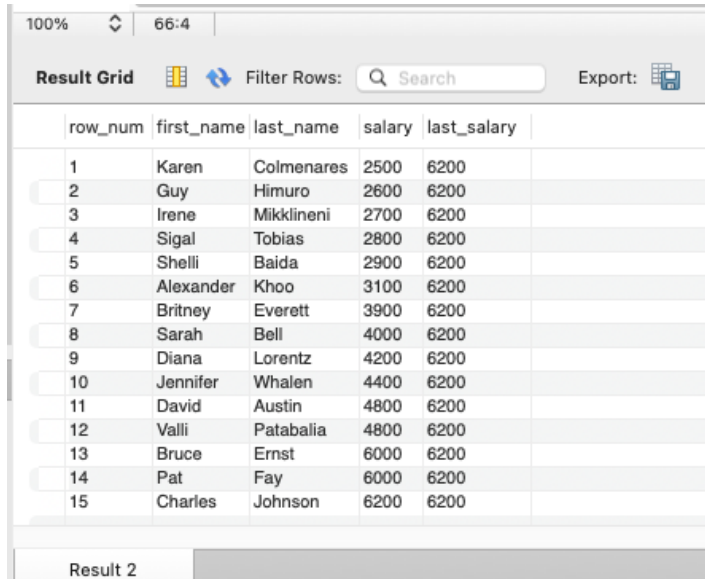
| row_num | first_name | last_name   | salary | first_salary |
|---------|------------|-------------|--------|--------------|
| 15      | Charles    | Johnson     | 6200   | 2500         |
| 13      | Bruce      | Ernst       | 6000   | 2500         |
| 14      | Pat        | Fay         | 6000   | 2500         |
| 11      | David      | Austin      | 4800   | 2500         |
| 12      | Valli      | Patabalia   | 4800   | 2500         |
| 10      | Jennifer   | Whalen      | 4400   | 2500         |
| 9       | Diana      | Lorentz     | 4200   | 2500         |
| 8       | Sarah      | Bell        | 4000   | 2500         |
| 7       | Britney    | Everett     | 3900   | 2500         |
| 6       | Alexander  | Khoo        | 3100   | 2500         |
| 5       | Shelli     | Baida       | 2900   | 2500         |
| 4       | Sigal      | Tobias      | 2800   | 2500         |
| 3       | Irene      | Mikkilineni | 2700   | 2500         |
| 2       | Guy        | Himuro      | 2600   | 2500         |
| 1       | Karen      | Colmena...  | 2500   | 2500         |

2. Write a query to compute for the **LAST\_VALUE()** and return the value along with the entire row.

#QUESTION 2

```
SELECT *,
    LAST_VALUE(salary) OVER (ORDER BY salary ASC ROWS BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED
    FOLLOWING) AS last_salary
FROM
    employees
ORDER BY
    salary ASC;
```

Capture of the result



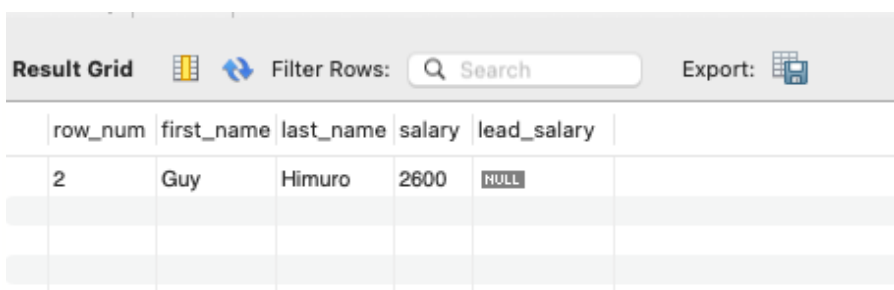
| row_num | first_name | last_name   | salary | last_salary |
|---------|------------|-------------|--------|-------------|
| 1       | Karen      | Colmenares  | 2500   | 6200        |
| 2       | Guy        | Himuro      | 2600   | 6200        |
| 3       | Irene      | Mikkilineni | 2700   | 6200        |
| 4       | Sigal      | Tobias      | 2800   | 6200        |
| 5       | Shelli     | Baida       | 2900   | 6200        |
| 6       | Alexander  | Khoo        | 3100   | 6200        |
| 7       | Britney    | Everett     | 3900   | 6200        |
| 8       | Sarah      | Bell        | 4000   | 6200        |
| 9       | Diana      | Lorentz     | 4200   | 6200        |
| 10      | Jennifer   | Whalen      | 4400   | 6200        |
| 11      | David      | Austin      | 4800   | 6200        |
| 12      | Valli      | Patabalia   | 4800   | 6200        |
| 13      | Bruce      | Ernst       | 6000   | 6200        |
| 14      | Pat        | Fay         | 6000   | 6200        |
| 15      | Charles    | Johnson     | 6200   | 6200        |

3. Write a query to compute for **LEAD(2)** for Guy and return the value along with the Guy's row.

#QUESTION 3

```
SELECT *,
    LEAD(salary, 2) OVER (ORDER BY salary ASC) AS lead_salary
FROM
    employees
WHERE
    first_name = 'Guy';
```

Capture of the result



| row_num | first_name | last_name | salary | lead_salary |
|---------|------------|-----------|--------|-------------|
| 2       | Guy        | Himuro    | 2600   | NULL        |

4. Write a query to compute for **LAG(4)** for Pat and return value along with Pat's row.

#QUESTION 4

```
SELECT *,  
    LAG(salary, 4) OVER (ORDER BY salary ASC) AS lag_salary  
FROM  
    employees  
WHERE  
    first_name = 'Pat';
```

Capture of the result

| Result Grid   Filter Rows: <input type="text" value="Search"/> Export:  |            |           |        |            |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|--------|------------|--|--|
| row_num                                                                                                                                                                                                                                                                                                                     | first_name | last_name | salary | lag_salary |  |  |
| 14                                                                                                                                                                                                                                                                                                                          | Pat        | Fay       | 6000   | NULL       |  |  |
|                                                                                                                                                                                                                                                                                                                             |            |           |        |            |  |  |
|                                                                                                                                                                                                                                                                                                                             |            |           |        |            |  |  |

5. Write a query to compute the **RANK()** and **DENSE\_RANK()** and return the entire dataset, including the rank and dense rank for each employee.

#QUESTION 5

```
SELECT *,  
    RANK() OVER (ORDER BY salary) AS nor_rank,  
    DENSE_RANK() OVER (ORDER BY salary) AS denserank  
FROM  
    employees  
ORDER BY  
    salary ASC;
```

Capture of the result

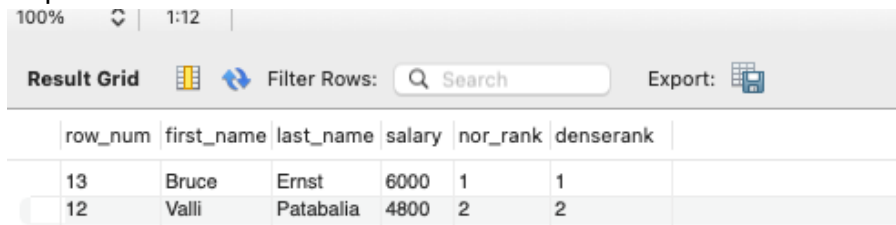
| Result Grid   Filter Rows: <input type="text" value="Search"/> Export:  |            |             |        |          |           |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|--------|----------|-----------|--|
| row_num                                                                                                                                                                                                                                                                                                                          | first_name | last_name   | salary | nor_rank | denserank |  |
| 1                                                                                                                                                                                                                                                                                                                                | Karen      | Colmenares  | 2500   | 1        | 1         |  |
| 2                                                                                                                                                                                                                                                                                                                                | Guy        | Himuro      | 2600   | 2        | 2         |  |
| 3                                                                                                                                                                                                                                                                                                                                | Irene      | Mikkilineni | 2700   | 3        | 3         |  |
| 4                                                                                                                                                                                                                                                                                                                                | Sigal      | Tobias      | 2800   | 4        | 4         |  |
| 5                                                                                                                                                                                                                                                                                                                                | Shelli     | Baida       | 2900   | 5        | 5         |  |
| 6                                                                                                                                                                                                                                                                                                                                | Alexander  | Khoo        | 3100   | 6        | 6         |  |
| 7                                                                                                                                                                                                                                                                                                                                | Britney    | Everett     | 3900   | 7        | 7         |  |
| 8                                                                                                                                                                                                                                                                                                                                | Sarah      | Bell        | 4000   | 8        | 8         |  |
| 9                                                                                                                                                                                                                                                                                                                                | Diana      | Lorentz     | 4200   | 9        | 9         |  |
| 10                                                                                                                                                                                                                                                                                                                               | Jennifer   | Whalen      | 4400   | 10       | 10        |  |
| 11                                                                                                                                                                                                                                                                                                                               | David      | Austin      | 4800   | 11       | 11        |  |
| 12                                                                                                                                                                                                                                                                                                                               | Valli      | Patabalia   | 4800   | 11       | 11        |  |
| 13                                                                                                                                                                                                                                                                                                                               | Bruce      | Ernst       | 6000   | 13       | 12        |  |
| 14                                                                                                                                                                                                                                                                                                                               | Pat        | Fay         | 6000   | 13       | 12        |  |
| 15                                                                                                                                                                                                                                                                                                                               | Charles    | Johnson     | 6200   | 15       | 13        |  |

6. Write a query to compute the **RANK()** and **DENSE\_RANK()** but only return Valli's and Bruce's rank and dense rank.

#QUESTION 6

```
SELECT *,  
       RANK() OVER (ORDER BY salary DESC) AS nor_rank,  
       DENSE_RANK() OVER (ORDER BY salary DESC) AS denserank  
FROM  
     employees  
WHERE first_name IN ('Valli', 'Bruce');
```

Capture of the result



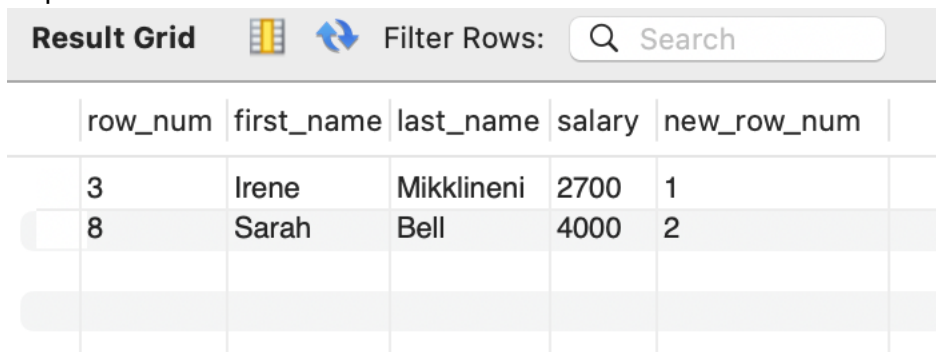
| row_num | first_name | last_name | salary | nor_rank | denserank |
|---------|------------|-----------|--------|----------|-----------|
| 13      | Bruce      | Ernst     | 6000   | 1        | 1         |
| 12      | Valli      | Patabalia | 4800   | 2        | 2         |

7. Write a query to compute the **ROW\_NUMBER()** for Irene and Sarah and only return the rows corresponding to them.

#QUESTION 7

```
SELECT *,  
       ROW_NUMBER() OVER (ORDER BY salary ASC) AS new_row_num  
FROM  
     employees  
WHERE first_name IN ('Irene', 'Sarah');
```

Capture of the result



| row_num | first_name | last_name | salary | new_row_num |
|---------|------------|-----------|--------|-------------|
| 3       | Irene      | Mikkineni | 2700   | 1           |
| 8       | Sarah      | Bell      | 4000   | 2           |

8. Write a query to compute the **PERCENT\_RANK()** and return the entire dataset, including the percent rank for each employee. Format your PERCENT\_RANK() values to 100%.

#QUESTION 8

```
SELECT *,
RANK() OVER (ORDER BY salary) AS s_rank,
CONCAT(FORMAT(PERCENT_RANK() OVER (ORDER BY salary) * 100, 2), '%') AS percentrank
FROM
employees;
```

Capture of the result

| Result Grid |         | Filter Rows: |             | Search | Export: |             |
|-------------|---------|--------------|-------------|--------|---------|-------------|
|             | row_num | first_name   | last_name   | salary | s_rank  | percentrank |
|             | 1       | Karen        | Colmenares  | 2500   | 1       | 0.00%       |
|             | 2       | Guy          | Himuro      | 2600   | 2       | 7.14%       |
|             | 3       | Irene        | Mikkilineni | 2700   | 3       | 14.29%      |
|             | 4       | Sigal        | Tobias      | 2800   | 4       | 21.43%      |
|             | 5       | Shelli       | Baida       | 2900   | 5       | 28.57%      |
|             | 6       | Alexander    | Khoo        | 3100   | 6       | 35.71%      |
|             | 7       | Britney      | Everett     | 3900   | 7       | 42.86%      |
|             | 8       | Sarah        | Bell        | 4000   | 8       | 50.00%      |
|             | 9       | Diana        | Lorentz     | 4200   | 9       | 57.14%      |
|             | 10      | Jennifer     | Whalen      | 4400   | 10      | 64.29%      |
|             | 11      | David        | Austin      | 4800   | 11      | 71.43%      |
|             | 12      | Valli        | Patabalia   | 4800   | 11      | 71.43%      |
|             | 13      | Bruce        | Ernst       | 6000   | 13      | 85.71%      |
|             | 14      | Pat          | Fay         | 6000   | 13      | 85.71%      |
|             | 15      | Charles      | Johnson     | 6200   | 15      | 100.00%     |

9. Write a query to compute the **CUME\_DIST()** and return the entire dataset, including the percentage rank for each employee. Format your CUME\_DIST() values to 2 decimal places.

#QUESTION 9

```
SELECT *,
CONCAT(FORMAT(PERCENT_RANK() OVER (ORDER BY salary) * 100, 2), '%') AS percentrank,
ROUND(CUME_DIST() OVER (ORDER BY salary ASC), 2) AS cumedist
FROM
employees
ORDER BY
salary ASC;
```

Capture of the result

| Result Grid   Filter Rows: <input type="text" value="Search"/> Export:  |         |            |            |        |             |          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------------|------------|--------|-------------|----------|
|                                                                                                                                                                                                                                                                                                                             | row_num | first_name | last_name  | salary | percentrank | cumedist |
|                                                                                                                                                                                                                                                                                                                             | 1       | Karen      | Colmenares | 2500   | 0.00%       | 0.07     |
|                                                                                                                                                                                                                                                                                                                             | 2       | Guy        | Himuro     | 2600   | 7.14%       | 0.13     |
|                                                                                                                                                                                                                                                                                                                             | 3       | Irene      | Mikklineni | 2700   | 14.29%      | 0.2      |
|                                                                                                                                                                                                                                                                                                                             | 4       | Sigal      | Tobias     | 2800   | 21.43%      | 0.27     |
|                                                                                                                                                                                                                                                                                                                             | 5       | Shelli     | Baida      | 2900   | 28.57%      | 0.33     |
|                                                                                                                                                                                                                                                                                                                             | 6       | Alexander  | Khoo       | 3100   | 35.71%      | 0.4      |
|                                                                                                                                                                                                                                                                                                                             | 7       | Britney    | Everett    | 3900   | 42.86%      | 0.47     |
|                                                                                                                                                                                                                                                                                                                             | 8       | Sarah      | Bell       | 4000   | 50.00%      | 0.53     |
|                                                                                                                                                                                                                                                                                                                             | 9       | Diana      | Lorentz    | 4200   | 57.14%      | 0.6      |
|                                                                                                                                                                                                                                                                                                                             | 10      | Jennifer   | Whalen     | 4400   | 64.29%      | 0.67     |
|                                                                                                                                                                                                                                                                                                                             | 11      | David      | Austin     | 4800   | 71.43%      | 0.8      |
|                                                                                                                                                                                                                                                                                                                             | 12      | Valli      | Patabalia  | 4800   | 71.43%      | 0.8      |
|                                                                                                                                                                                                                                                                                                                             | 13      | Bruce      | Ernst      | 6000   | 85.71%      | 0.93     |
|                                                                                                                                                                                                                                                                                                                             | 14      | Pat        | Fay        | 6000   | 85.71%      | 0.93     |
|                                                                                                                                                                                                                                                                                                                             | 15      | Charles    | Johnson    | 6200   | 100.00%     | 1        |

10. Write a query to compute the **NTILE(4)** and return the entire dataset showing approximately equal groups/buckets.

#QUESTION 10

```
SELECT *,
       NTILE(4) OVER (ORDER BY salary ASC) AS '4ntile'
FROM
  Employees
```

Capture of the result

| Result Grid   Filter Rows: <input type="text" value="Search"/> |         |            |            |        |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------------|------------|--------|--------|
|                                                                                                                                                                                                                                      | row_num | first_name | last_name  | salary | 4ntile |
|                                                                                                                                                                                                                                      | 1       | Karen      | Colmenares | 2500   | 1      |
|                                                                                                                                                                                                                                      | 2       | Guy        | Himuro     | 2600   | 1      |
|                                                                                                                                                                                                                                      | 3       | Irene      | Mikklineni | 2700   | 1      |
|                                                                                                                                                                                                                                      | 4       | Sigal      | Tobias     | 2800   | 1      |
|                                                                                                                                                                                                                                      | 5       | Shelli     | Baida      | 2900   | 2      |
|                                                                                                                                                                                                                                      | 6       | Alexander  | Khoo       | 3100   | 2      |
|                                                                                                                                                                                                                                      | 7       | Britney    | Everett    | 3900   | 2      |
|                                                                                                                                                                                                                                      | 8       | Sarah      | Bell       | 4000   | 2      |
|                                                                                                                                                                                                                                      | 9       | Diana      | Lorentz    | 4200   | 3      |
|                                                                                                                                                                                                                                      | 10      | Jennifer   | Whalen     | 4400   | 3      |
|                                                                                                                                                                                                                                      | 11      | David      | Austin     | 4800   | 3      |
|                                                                                                                                                                                                                                      | 12      | Valli      | Patabalia  | 4800   | 3      |
|                                                                                                                                                                                                                                      | 13      | Bruce      | Ernst      | 6000   | 4      |
|                                                                                                                                                                                                                                      | 14      | Pat        | Fay        | 6000   | 4      |
|                                                                                                                                                                                                                                      | 15      | Charles    | Johnson    | 6200   | 4      |

## Windowing activity

- Find a running total of covid admissions over three days (i.e., previous day, current day and next day)

#Windowing activity 1

```
SELECT
  date,
  admissions,
  SUM(admissions) OVER (ORDER BY date ROWS BETWEEN 1 PRECEDING AND 1 FOLLOWING) AS cumOverAdmissions
FROM covid_statistics
ORDER BY date;
```

### Capture of the result

| Result Grid   Filter Rows: <input type="text" value="Search"/> |            |            |                    |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------|--------------------|--|
|                                                                                                                                                                                                                                  | date       | admissions | cumOverAdmissio... |  |
|                                                                                                                                                                                                                                  | 2020-05-24 | 3046       | 6262               |  |
|                                                                                                                                                                                                                                  | 2020-05-25 | 3216       | 9701               |  |
|                                                                                                                                                                                                                                  | 2020-05-26 | 3439       | 10457              |  |
|                                                                                                                                                                                                                                  | 2020-05-27 | 3802       | 11386              |  |
|                                                                                                                                                                                                                                  | 2020-05-28 | 4145       | 12037              |  |
|                                                                                                                                                                                                                                  | 2020-05-29 | 4090       | 12596              |  |
|                                                                                                                                                                                                                                  | 2020-05-30 | 4361       | 12997              |  |
|                                                                                                                                                                                                                                  | 2020-05-31 | 4546       | 13625              |  |
|                                                                                                                                                                                                                                  | 2020-06-01 | 4718       | 14191              |  |
|                                                                                                                                                                                                                                  | 2020-06-02 | 4927       | 14723              |  |
|                                                                                                                                                                                                                                  | 2020-06-03 | 5078       | 15382              |  |
|                                                                                                                                                                                                                                  | 2020-06-04 | 5377       | 16400              |  |
|                                                                                                                                                                                                                                  | 2020-06-06 | 5945       | 17502              |  |
|                                                                                                                                                                                                                                  | 2020-06-07 | 6180       | 18478              |  |
|                                                                                                                                                                                                                                  | 2020-06-08 | 6353       | 19215              |  |
|                                                                                                                                                                                                                                  | 2020-06-09 | 6682       | 20027              |  |
|                                                                                                                                                                                                                                  | 2020-06-10 | 6992       | 20965              |  |
|                                                                                                                                                                                                                                  | 2020-06-11 | 7291       | 21839              |  |
|                                                                                                                                                                                                                                  | 2020-06-12 | 7556       | 22684              |  |
|                                                                                                                                                                                                                                  | 2020-06-13 | 7837       | 23446              |  |
|                                                                                                                                                                                                                                  | 2020-06-14 | 8053       | 24182              |  |
|                                                                                                                                                                                                                                  | 2020-06-15 | 8292       | 25033              |  |
|                                                                                                                                                                                                                                  | 2020-06-16 | 8688       | 25882              |  |
|                                                                                                                                                                                                                                  | 2020-06-17 | 8902       | 26911              |  |
|                                                                                                                                                                                                                                  | 2020-06-18 | 9321       | 27916              |  |
|                                                                                                                                                                                                                                  | 2020-06-19 | 9693       | 29034              |  |
|                                                                                                                                                                                                                                  | 2020-06-20 | 10020      | 30073              |  |
|                                                                                                                                                                                                                                  | 2020-06-21 | 10360      | 31080              |  |
|                                                                                                                                                                                                                                  | 2020-06-22 | 10700      | 32380              |  |
|                                                                                                                                                                                                                                  | 2020-06-23 | 11320      | 33870              |  |
|                                                                                                                                                                                                                                  | 2020-06-24 | 11850      | 35430              |  |
|                                                                                                                                                                                                                                  | 2020-06-25 | 12260      | 37960              |  |
|                                                                                                                                                                                                                                  | 2020-06-27 | 13850      | 40210              |  |
|                                                                                                                                                                                                                                  | 2020-06-28 | 14100      | 42510              |  |
|                                                                                                                                                                                                                                  | 2020-06-29 | 14560      | 43790              |  |
|                                                                                                                                                                                                                                  | 2020-06-30 | 15130      | 45060              |  |
|                                                                                                                                                                                                                                  | 2020-07-01 | 15370      | 46360              |  |
|                                                                                                                                                                                                                                  | 2020-07-02 | 15860      | 47970              |  |
|                                                                                                                                                                                                                                  | 2020-07-03 | 16740      | 32600              |  |
|                                                                                                                                                                                                                                  |            |            |                    |  |
|                                                                                                                                                                                                                                  |            |            |                    |  |



- Find a 7-day moving average of covid patients in ICU.

```
SELECT
  date,
  ICU,
  ROUND(AVG(ICU) OVER (ORDER BY date ROWS BETWEEN 3 PRECEDING AND 3 FOLLOWING),1) AS avg_7_days
FROM covid_statistics
ORDER BY date;
```

Capture of the result

| Result Grid   Filter Rows: <input type="text" value="Search"/> <span>Export</span> |            |     |            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|------------|
|                                                                                                                                                                                                                                                      | date       | ICU | avg_7_days |
|                                                                                                                                                                                                                                                      | 2020-05-24 | 151 | 152.5      |
|                                                                                                                                                                                                                                                      | 2020-05-25 | 146 | 161.6      |
|                                                                                                                                                                                                                                                      | 2020-05-26 | 151 | 166.0      |
|                                                                                                                                                                                                                                                      | 2020-05-27 | 162 | 171.6      |
|                                                                                                                                                                                                                                                      | 2020-05-28 | 198 | 179.0      |
|                                                                                                                                                                                                                                                      | 2020-05-29 | 188 | 187.9      |
|                                                                                                                                                                                                                                                      | 2020-05-30 | 205 | 197.3      |
|                                                                                                                                                                                                                                                      | 2020-05-31 | 203 | 205.4      |
|                                                                                                                                                                                                                                                      | 2020-06-01 | 208 | 211.7      |
|                                                                                                                                                                                                                                                      | 2020-06-02 | 217 | 222.7      |
|                                                                                                                                                                                                                                                      | 2020-06-03 | 219 | 231.7      |
|                                                                                                                                                                                                                                                      | 2020-06-04 | 242 | 245.6      |
|                                                                                                                                                                                                                                                      | 2020-06-06 | 265 | 256.7      |
|                                                                                                                                                                                                                                                      | 2020-06-07 | 268 | 268.7      |
|                                                                                                                                                                                                                                                      | 2020-06-08 | 300 | 280.3      |
|                                                                                                                                                                                                                                                      | 2020-06-09 | 286 | 292.3      |
|                                                                                                                                                                                                                                                      | 2020-06-10 | 301 | 301.4      |
|                                                                                                                                                                                                                                                      | 2020-06-11 | 300 | 311.4      |
|                                                                                                                                                                                                                                                      | 2020-06-12 | 326 | 317.1      |
|                                                                                                                                                                                                                                                      | 2020-06-13 | 329 | 326.6      |
|                                                                                                                                                                                                                                                      | 2020-06-14 | 338 | 337.9      |
|                                                                                                                                                                                                                                                      | 2020-06-15 | 340 | 351.9      |
|                                                                                                                                                                                                                                                      | 2020-06-16 | 352 | 368.6      |
|                                                                                                                                                                                                                                                      | 2020-06-17 | 380 | 388.4      |
|                                                                                                                                                                                                                                                      | 2020-06-18 | 398 | 410.1      |
|                                                                                                                                                                                                                                                      | 2020-06-19 | 443 | 432.9      |
|                                                                                                                                                                                                                                                      | 2020-06-20 | 468 | 459.4      |
|                                                                                                                                                                                                                                                      | 2020-06-21 | 490 | 483.0      |
|                                                                                                                                                                                                                                                      | 2020-06-22 | 499 | 507.7      |
|                                                                                                                                                                                                                                                      | 2020-06-23 | 538 | 536.6      |
|                                                                                                                                                                                                                                                      | 2020-06-24 | 545 | 565.1      |
|                                                                                                                                                                                                                                                      | 2020-06-25 | 571 | 590.4      |
|                                                                                                                                                                                                                                                      | 2020-06-27 | 645 | 617.7      |
|                                                                                                                                                                                                                                                      | 2020-06-28 | 668 | 640.7      |
|                                                                                                                                                                                                                                                      | 2020-06-29 | 667 | 664.0      |
|                                                                                                                                                                                                                                                      | 2020-06-30 | 690 | 693.0      |
|                                                                                                                                                                                                                                                      | 2020-07-01 | 699 | 701.0      |
|                                                                                                                                                                                                                                                      | 2020-07-02 | 708 | 707.6      |
|                                                                                                                                                                                                                                                      | 2020-07-03 | 774 | 717.8      |
|                                                                                                                                                                                                                                                      |            |     |            |
|                                                                                                                                                                                                                                                      |            |     |            |



- Find a running total of covid deaths in each state.

#Windowing activity 3

```
SELECT
  date,
  state,
  hospital_deaths AS deaths,
  SUM(hospital_deaths) OVER (PARTITION BY state ORDER BY date) AS total_deaths
FROM covid_statistics
ORDER BY state, date;
```

### Capture of the result

| Result Grid   Filter Rows: <input type="text" value="Search"/> |            |       |        |              |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|--------|--------------|--|
|                                                                                                                                                                                                                                  | date       | state | deaths | total_deaths |  |
|                                                                                                                                                                                                                                  | 2020-05-24 | AL    | 389    | 389          |  |
|                                                                                                                                                                                                                                  | 2020-06-01 | AL    | 608    | 997          |  |
|                                                                                                                                                                                                                                  | 2020-06-02 | AL    | 643    | 1640         |  |
|                                                                                                                                                                                                                                  | 2020-06-13 | AL    | 1088   | 2728         |  |
|                                                                                                                                                                                                                                  | 2020-06-17 | AL    | 1265   | 3993         |  |
|                                                                                                                                                                                                                                  | 2020-05-25 | AZ    | 409    | 409          |  |
|                                                                                                                                                                                                                                  | 2020-06-03 | AZ    | 663    | 1072         |  |
|                                                                                                                                                                                                                                  | 2020-06-11 | AZ    | 1016   | 2088         |  |
|                                                                                                                                                                                                                                  | 2020-06-14 | AZ    | 1128   | 3216         |  |
|                                                                                                                                                                                                                                  | 2020-06-18 | AZ    | 1333   | 4549         |  |
|                                                                                                                                                                                                                                  | 2020-06-23 | AZ    | 1592   | 6141         |  |
|                                                                                                                                                                                                                                  | 2020-06-25 | AZ    | 1739   | 7880         |  |
|                                                                                                                                                                                                                                  | 2020-05-27 | DC    | 472    | 472          |  |
|                                                                                                                                                                                                                                  | 2020-05-29 | DC    | 519    | 991          |  |
|                                                                                                                                                                                                                                  | 2020-05-31 | DC    | 587    | 1578         |  |
|                                                                                                                                                                                                                                  | 2020-06-06 | DC    | 775    | 2353         |  |
|                                                                                                                                                                                                                                  | 2020-06-08 | DC    | 854    | 3207         |  |
|                                                                                                                                                                                                                                  | 2020-06-10 | DC    | 960    | 4167         |  |
|                                                                                                                                                                                                                                  | 2020-06-16 | DC    | 1226   | 5393         |  |
|                                                                                                                                                                                                                                  | 2020-06-20 | DC    | 1420   | 6813         |  |
|                                                                                                                                                                                                                                  | 2020-06-22 | DC    | 1515   | 8328         |  |
|                                                                                                                                                                                                                                  | 2020-06-30 | DC    | 2189   | 10517        |  |
|                                                                                                                                                                                                                                  | 2020-07-03 | DC    | 2400   | 12917        |  |
|                                                                                                                                                                                                                                  | 2020-05-26 | DE    | 447    | 447          |  |
|                                                                                                                                                                                                                                  | 2020-06-04 | DE    | 702    | 1149         |  |
|                                                                                                                                                                                                                                  | 2020-06-07 | DE    | 825    | 1974         |  |
|                                                                                                                                                                                                                                  | 2020-06-09 | DE    | 921    | 2895         |  |
|                                                                                                                                                                                                                                  | 2020-06-12 | DE    | 1035   | 3930         |  |
|                                                                                                                                                                                                                                  | 2020-06-15 | DE    | 1156   | 5086         |  |
|                                                                                                                                                                                                                                  | 2020-06-19 | DE    | 1372   | 6458         |  |
|                                                                                                                                                                                                                                  | 2020-06-21 | DE    | 1484   | 7942         |  |
|                                                                                                                                                                                                                                  | 2020-06-27 | DE    | 1966   | 9908         |  |
|                                                                                                                                                                                                                                  | 2020-06-29 | DE    | 2116   | 12024        |  |
|                                                                                                                                                                                                                                  | 2020-05-28 | FL    | 497    | 497          |  |
|                                                                                                                                                                                                                                  | 2020-05-30 | FL    | 559    | 1056         |  |
|                                                                                                                                                                                                                                  | 2020-06-24 | FL    | 1678   | 2734         |  |
|                                                                                                                                                                                                                                  | 2020-06-28 | FL    | 1995   | 4729         |  |
|                                                                                                                                                                                                                                  | 2020-07-01 | FL    | 2217   | 6946         |  |
|                                                                                                                                                                                                                                  | 2020-07-02 | FL    | 2273   | 9219         |  |
|                                                                                                                                                                                                                                  |            |       |        |              |  |
|                                                                                                                                                                                                                                  |            |       |        |              |  |

## PART II: OLAP queries

- What are the total sales of each product in each region and country?

#OLAP QUERY 1

SELECT

product,

region,

country,

SUM(sales) AS Total\_Sales

FROM

regional\_sales

GROUP BY

product,

region,

country

ORDER BY

product,

Total\_Sales DESC;

Capture of the result

| Result Grid   Filter Rows: <input type="text" value="Search"/> |         |               |             |             |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------------|-------------|-------------|--|
|                                                                                                                                                                                                                                  | product | region        | country     | Total_Sales |  |
|                                                                                                                                                                                                                                  | A       | North America | USA         | 8200        |  |
|                                                                                                                                                                                                                                  | A       | Asia          | China       | 4200        |  |
|                                                                                                                                                                                                                                  | A       | North America | Canada      | 3200        |  |
|                                                                                                                                                                                                                                  | A       | Asia          | Japan       | 800         |  |
|                                                                                                                                                                                                                                  | A       | Europe        | Germany     | 300         |  |
|                                                                                                                                                                                                                                  | B       | Europe        | France      | 8700        |  |
|                                                                                                                                                                                                                                  | B       | Australia     | Australia   | 6000        |  |
|                                                                                                                                                                                                                                  | B       | Australia     | New Zealand | 5500        |  |
|                                                                                                                                                                                                                                  | B       | Europe        | Germany     | 700         |  |
|                                                                                                                                                                                                                                  | B       | Europe        | Italy       | 400         |  |

- What are the total sales of each product in each region and country, including subtotals and the grand total?

#OLAP QUERY 2

SELECT

product,

region,

country,

SUM(sales) AS Total\_Sales

FROM

regional\_sales

GROUP BY

product,

region,

country WITH ROLLUP;

## Capture of the result

| Result Grid   Filter Rows: <input type="text" value="Search"/> Export:  |         |               |             |             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------------|-------------|-------------|
|                                                                                                                                                                                                                                                                                                                            | product | region        | country     | Total_Sales |
|                                                                                                                                                                                                                                                                                                                            | A       | Asia          | China       | 4200        |
|                                                                                                                                                                                                                                                                                                                            | A       | Asia          | Japan       | 800         |
|                                                                                                                                                                                                                                                                                                                            | A       | Asia          | NULL        | 5000        |
|                                                                                                                                                                                                                                                                                                                            | A       | Europe        | Germany     | 300         |
|                                                                                                                                                                                                                                                                                                                            | A       | Europe        | NULL        | 300         |
|                                                                                                                                                                                                                                                                                                                            | A       | North America | Canada      | 3200        |
|                                                                                                                                                                                                                                                                                                                            | A       | North America | USA         | 8200        |
|                                                                                                                                                                                                                                                                                                                            | A       | North America | NULL        | 11400       |
|                                                                                                                                                                                                                                                                                                                            | A       | NULL          | NULL        | 16700       |
|                                                                                                                                                                                                                                                                                                                            | B       | Australia     | Australia   | 6000        |
|                                                                                                                                                                                                                                                                                                                            | B       | Australia     | New Zealand | 5500        |
|                                                                                                                                                                                                                                                                                                                            | B       | Australia     | NULL        | 11500       |
|                                                                                                                                                                                                                                                                                                                            | B       | Europe        | France      | 8700        |
|                                                                                                                                                                                                                                                                                                                            | B       | Europe        | Germany     | 700         |
|                                                                                                                                                                                                                                                                                                                            | B       | Europe        | Italy       | 400         |
|                                                                                                                                                                                                                                                                                                                            | B       | Europe        | NULL        | 9800        |
|                                                                                                                                                                                                                                                                                                                            | B       | NULL          | NULL        | 21300       |
|                                                                                                                                                                                                                                                                                                                            | NULL    | NULL          | NULL        | 38000       |

- What are the total sales generated by each product in each region and country, including all subtotals and grand totals?

#OLAP QUERY 3

SELECT

COALESCE(product, 'All Products') AS product,  
COALESCE(region, 'All Regions') AS region,  
COALESCE(country, 'All Countries') AS country,  
SUM(sales) AS total\_sales

FROM

regional\_sales

GROUP BY

product, region, country

WITH ROLLUP;

## Capture of the result

| Result Grid   Filter Rows: <input type="text" value="Search"/> Export:  |              |               |               |             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------|---------------|-------------|
|                                                                                                                                                                                                                                                                                                                                  | product      | region        | country       | total_sales |
|                                                                                                                                                                                                                                                                                                                                  | A            | Asia          | China         | 4200        |
|                                                                                                                                                                                                                                                                                                                                  | A            | Asia          | Japan         | 800         |
|                                                                                                                                                                                                                                                                                                                                  | A            | Asia          | All Countries | 5000        |
|                                                                                                                                                                                                                                                                                                                                  | A            | Europe        | Germany       | 300         |
|                                                                                                                                                                                                                                                                                                                                  | A            | Europe        | All Countries | 300         |
|                                                                                                                                                                                                                                                                                                                                  | A            | North America | Canada        | 3200        |
|                                                                                                                                                                                                                                                                                                                                  | A            | North America | USA           | 8200        |
|                                                                                                                                                                                                                                                                                                                                  | A            | North America | All Countries | 11400       |
|                                                                                                                                                                                                                                                                                                                                  | A            | All Regions   | All Countries | 16700       |
|                                                                                                                                                                                                                                                                                                                                  | B            | Australia     | Australia     | 6000        |
|                                                                                                                                                                                                                                                                                                                                  | B            | Australia     | New Zealand   | 5500        |
|                                                                                                                                                                                                                                                                                                                                  | B            | Australia     | All Countries | 11500       |
|                                                                                                                                                                                                                                                                                                                                  | B            | Europe        | France        | 8700        |
|                                                                                                                                                                                                                                                                                                                                  | B            | Europe        | Germany       | 700         |
|                                                                                                                                                                                                                                                                                                                                  | B            | Europe        | Italy         | 400         |
|                                                                                                                                                                                                                                                                                                                                  | B            | Europe        | All Countries | 9800        |
|                                                                                                                                                                                                                                                                                                                                  | B            | All Regions   | All Countries | 21300       |
|                                                                                                                                                                                                                                                                                                                                  | All Products | All Regions   | All Countries | 38000       |

Material required: regional sales dataset attached.

**Submission requirements:**

- the query statements.
- the SQL commands.
- query result/output (screenshots).
- Submit as a single PDF file via the submission link on the Blackboard.

Deadline: 6<sup>th</sup> July 2024 Time: 23h59 (Chicago Time).

```
-- Check if the database 'Homework4DB' exists, and create it if it does not
CREATE DATABASE IF NOT EXISTS Homework4DB2;
```

```
USE Homework4DB2;
```

```
-----
-- COVID STATISTICS TABLE CREATION & DATA INSERTION
-----
```

```
CREATE TABLE covid_statistics (
  date DATE,
  yyymmdd INT,
  state VARCHAR(2),
  admissions INT,
  current_num_in_hospital INT,
  general INT,
  high_care INT,
  ICU INT,
  isolation INT,
  total_healthcare_workers_admitted INT,
  num_discharged_alive INT,
  hospital_deaths INT
);
```

```
INSERT INTO covid_statistics (
  date, yyymmdd, state, admissions, current_num_in_hospital, general, high_care, ICU, isolation,
  total_healthcare_workers_admitted, num_discharged_alive, hospital_deaths
) VALUES
('2020-05-24', 20200524, 'AL', 3046, 991, 715, 94, 151, 31, 165, 1615, 389),
('2020-05-25', 20200525, 'AZ', 3216, 1086, 805, 100, 146, 35, 156, 1669, 409),
('2020-05-26', 20200526, 'DE', 3439, 1139, 848, 96, 151, 44, 164, 1800, 447),
('2020-05-27', 20200527, 'DC', 3802, 1364, 1042, 119, 162, 41, 180, 1910, 472),
('2020-05-28', 20200528, 'FL', 4145, 1581, 1218, 119, 198, 46, 187, 2011, 497),
('2020-05-29', 20200529, 'DC', 4090, 1395, 1046, 117, 188, 44, 182, 2118, 519),
('2020-05-30', 20200530, 'FL', 4361, 1458, 1098, 115, 205, 40, 187, 2285, 559),
('2020-05-31', 20200531, 'DC', 4546, 1555, 1204, 108, 203, 40, 188, 2346, 587),
('2020-06-01', 20200601, 'AL', 4718, 1607, 1255, 101, 208, 43, 189, 2441, 608),
('2020-06-02', 20200602, 'AL', 4927, 1659, 1283, 115, 217, 44, 193, 2562, 643),
('2020-06-03', 20200603, 'AZ', 5078, 1638, 1258, 114, 219, 47, 195, 2713, 663),
('2020-06-04', 20200604, 'DE', 5377, 1717, 1316, 115, 242, 44, 204, 2893, 702),
('2020-06-06', 20200606, 'DC', 5945, 1916, 1456, 138, 265, 57, 210, 3175, 775),
('2020-06-07', 20200607, 'DE', 6180, 1928, 1466, 133, 268, 61, 214, 3348, 825),
('2020-06-08', 20200608, 'DC', 6353, 1974, 1500, 100, 300, 100, 215, 3446, 854),
('2020-06-09', 20200609, 'DE', 6682, 2054, 1540, 159, 286, 69, 219, 3628, 921),
('2020-06-10', 20200610, 'DC', 6992, 2113, 1583, 157, 301, 72, 227, 3836, 960),
('2020-06-11', 20200611, 'AZ', 7291, 2140, 1601, 169, 300, 70, 230, 4052, 1016),
('2020-06-12', 20200612, 'DE', 7556, 2287, 1709, 174, 326, 78, 233, 4148, 1035),
('2020-06-13', 20200613, 'AL', 7837, 2263, 1684, 176, 329, 71, 237, 4396, 1088),
('2020-06-14', 20200614, 'AZ', 8053, 2332, 1739, 183, 338, 72, 238, 4502, 1128),
('2020-06-15', 20200615, 'DE', 8292, 2468, 1869, 189, 340, 70, 239, 4576, 1156),
('2020-06-16', 20200616, 'DC', 8688, 2565, 1922, 216, 352, 75, 245, 4804, 1226),
('2020-06-17', 20200617, 'AL', 8902, 2661, 1983, 223, 380, 75, 249, 4878, 1265),
('2020-06-18', 20200618, 'AZ', 9321, 2741, 2008, 249, 398, 86, 264, 5147, 1333),
('2020-06-19', 20200619, 'DE', 9693, 2949, 2164, 263, 443, 79, 265, 5257, 1372),
```

```
('2020-06-20', 20200620, 'DC', 10020, 3005, 2163, 292, 468, 82, 272, 5479, 1420),
('2020-06-21', 20200621, 'DE', 10360, 3088, 2199, 315, 490, 84, 278, 5669, 1484),
('2020-06-22', 20200622, 'DC', 10700, 3263, 2348, 333, 499, 83, 290, 5805, 1515),
('2020-06-23', 20200623, 'AZ', 11320, 3501, 2529, 341, 538, 93, 302, 6096, 1592),
('2020-06-24', 20200624, 'FL', 11850, 3622, 2619, 371, 545, 87, 314, 6397, 1678),
('2020-06-25', 20200625, 'AZ', 12260, 3567, 2526, 376, 571, 94, 317, 6807, 1739),
('2020-06-27', 20200627, 'DE', 13850, 4051, 2861, 412, 645, 124, 357, 7682, 1966),
('2020-06-28', 20200628, 'FL', 14100, 4150, 2941, 421, 668, 120, 365, 7797, 1995),
('2020-06-29', 20200629, 'DE', 14560, 4250, 3035, 420, 667, 128, 376, 8042, 2116),
('2020-06-30', 20200630, 'DC', 15130, 4418, 3160, 452, 690, 116, 394, 8364, 2189),
('2020-07-01', 20200701, 'FL', 15370, 4461, 3173, 452, 699, 137, 396, 8528, 2217),
('2020-07-02', 20200702, 'FL', 15860, 4616, 3313, 464, 708, 131, 402, 8805, 2273),
('2020-07-03', 20200703, 'DC', 16740, 5006, 3614, 471, 774, 147, 434, 9166, 2400);
```

---

## -- REGIONS TABLE CREATION & DATA INSERTION

---

```
create table regional_sales (
product char(100),
region char(100),
country char(100),
city char(100),
sales int)
;
```

```
INSERT INTO regional_sales (Product, Region, Country, City, Sales) VALUES
```

```
('A', 'Asia', 'China', 'Beijing', 2500),
('A', 'Asia', 'China', 'Beijing', 1700),
('A', 'Asia', 'Japan', 'Tokyo', 800),
('A', 'North America', 'Canada', 'Toronto', 900),
('A', 'North America', 'USA', 'New York', 400),
('A', 'North America', 'Canada', 'Toronto', 2300),
('A', 'North America', 'USA', 'Chicago', 7800),
('A', 'Europe', 'Germany', 'Berlin', 300),
('B', 'Europe', 'Germany', 'Berlin', 700),
('B', 'Europe', 'France', 'Paris', 5500),
('B', 'Europe', 'France', 'Paris', 2300),
('B', 'Europe', 'France', 'Paris', 900),
('B', 'Europe', 'Italy', 'Rome', 400),
('B', 'Australia', 'Australia', 'Canberra', 2500),
('B', 'Australia', 'Australia', 'Canberra', 3500),
('B', 'Australia', 'New Zealand', 'Wellington', 5500);
```

---

## -- EMPLOYEES TABLE CREATION & DATA INSERTION

---

```
CREATE TABLE employees (
row_num INT PRIMARY KEY,
first_name VARCHAR(50),
last_name VARCHAR(50),
salary int
```

```
);

INSERT INTO employees (row_num, first_name, last_name, salary) VALUES
(1, 'Karen', 'Colmenares', 2500),
(2, 'Guy', 'Himuro', 2600),
(3, 'Irene', 'Mikklineni', 2700),
(4, 'Sigal', 'Tobias', 2800),
(5, 'Shelli', 'Baida', 2900),
(6, 'Alexander', 'Khoo', 3100),
(7, 'Britney', 'Everett', 3900),
(8, 'Sarah', 'Bell', 4000),
(9, 'Diana', 'Lorentz', 4200),
(10, 'Jennifer', 'Whalen', 4400),
(11, 'David', 'Austin', 4800),
(12, 'Valli', 'Patabalia', 4800),
(13, 'Bruce', 'Ernst', 6000),
(14, 'Pat', 'Fay', 6000),
(15, 'Charles', 'Johnson', 6200);
```

---

## -- PART I. ADVANCED WINDOW FUNCTIONS

---

### #QUESTION 1

```
SELECT *,
    FIRST_VALUE(salary) OVER (ORDER BY salary ASC) AS first_salary
FROM
    employees
ORDER BY
    Salary DESC;
```

### #QUESTION 2

```
SELECT *,
    LAST_VALUE(salary) OVER (ORDER BY salary ASC ROWS BETWEEN UNBOUNDED
PRECEDING AND UNBOUNDED FOLLOWING) AS last_salary
FROM
    employees
ORDER BY
    salary ASC;
```

### #QUESTION 3

```
SELECT *,
    LEAD(salary, 2) OVER (ORDER BY salary ASC) AS lead_salary
FROM
    employees
WHERE
    first_name = 'Guy';
```

### #QUESTION 4

```
SELECT *,
    LAG(salary, 4) OVER (ORDER BY salary ASC) AS lag_salary
FROM
    employees
```

```
WHERE
    first_name = 'Pat';
```

#### #QUESTION 5

```
SELECT *,
    RANK() OVER (ORDER BY salary) AS nor_rank,
    DENSE_RANK() OVER (ORDER BY salary) AS denserank
FROM
    employees
ORDER BY
    salary ASC;
```

#### #QUESTION 6

```
SELECT *,
    RANK() OVER (ORDER BY salary DESC) AS nor_rank,
    DENSE_RANK() OVER (ORDER BY salary DESC) AS denserank
FROM
    employees
WHERE first_name IN ('Valli', 'Bruce');
```

#### #QUESTION 7

```
SELECT *,
    ROW_NUMBER() OVER (ORDER BY salary ASC) AS new_row_num
FROM
    employees
WHERE first_name IN ('Irene', 'Sarah');
```

#### #QUESTION 8

```
SELECT *,
    RANK() OVER (ORDER BY salary) AS s_rank,
    CONCAT(FORMAT(PERCENT_RANK() OVER (ORDER BY salary) * 100, 2), '%') AS percentrank
FROM
    employees;
```

#### #QUESTION 9

```
SELECT *,
    CONCAT(FORMAT(PERCENT_RANK() OVER (ORDER BY salary) * 100, 2), '%') AS percentrank,
    ROUND(CUME_DIST() OVER (ORDER BY salary ASC), 2) AS cumedist
FROM
    employees
ORDER BY
    salary ASC;
```

#### #QUESTION 10

```
SELECT *,
    NTILE(4) OVER (ORDER BY salary ASC) AS '4ntile'
FROM
    Employees;
```



---

## -- PART I. WINDOWING

---

#Windowing activity 1

```
SELECT
    date,
    admissions,
    SUM(admissions) OVER (ORDER BY date ROWS BETWEEN 1 PRECEDING AND 1
FOLLOWING) AS cumOverAdmissions
FROM covid_statistics
ORDER BY date;
```

#Windowing activity 2

```
SELECT
    date,
    ICU,
    ROUND(AVG(ICU) OVER (ORDER BY date ROWS BETWEEN 3 PRECEDING AND 3
FOLLOWING),1) AS avg_7_days
FROM covid_statistics
ORDER BY date;
```

#Windowing activity 3

```
SELECT
    date,
    state,
    hospital_deaths AS deaths,
    SUM(hospital_deaths) OVER (PARTITION BY state ORDER BY date) AS total_deaths
FROM covid_statistics
ORDER BY state, date;
```

---

## -- PART II. OLAP QUERIES

---

#OLAP QUERY 1

```
SELECT
    product,
    region,
    country,
    SUM(sales) AS Total_Sales
FROM
    regional_sales
GROUP BY
    product,
    region,
    country
ORDER BY
    product,
```

Total\_Sales DESC;

#OLAP QUERY 2

```
SELECT
    product,
    region,
    country,
    SUM(sales) AS Total_Sales
FROM
    regional_sales
GROUP BY
    product,
    region,
    country WITH ROLLUP;
```

#OLAP QUERY 3

```
SELECT
    COALESCE(product, 'All Products') AS product,
    COALESCE(region, 'All Regions') AS region,
    COALESCE(country, 'All Countries') AS country,
    SUM(sales) AS total_sales
FROM
    regional_sales
GROUP BY
    product, region, country
WITH ROLLUP;
```